Teacher Attendance Incentives and Recorded Student Departure

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Abstract

This paper replicates and extends selected findings from Duflo et al. (2012). I ask whether the effect of camera monitoring and attendance-linked pay on recorded departure differs according to children's baseline written-versus-oral test status. Using the original replication data, I reproduce the effects on school opening and three recorded-departure outcomes. I then estimate treatment effects for baseline writers and nonwriters. The effect on recorded departure is 26.1 percentage points for writers and 5.0 points for nonwriters. The 21.1-point difference remains statistically significant after adjustment across the three departure outcomes. The residual-departure interaction is positive, while the government-school-marker interaction is imprecise. However, recorded departure depends on checks when schools were open, which treatment itself affected. These exploratory results show different effects on recorded departure across baseline groups; they do not establish whether children left education or transferred.

1. Introduction

I ask whether the effect of teacher attendance incentives on leaving a nonformal education center differs according to whether children took the written or oral baseline test. Using publicly available replication files, I reproduce the estimated treatment effects on school opening in Table 2, Panel A, and the three recorded-departure outcomes in Table 10 of Duflo et al. (2012). I then extend this analysis by estimating treatment effects separately for baseline writers and nonwriters and testing the difference between them.

The original study is about Seva Mandir nonformal education centers in rural Rajasthan, India. The purpose of these centers was to offer basic education to children and prepare them for enrollment in government schools. They typically have only one teacher, highlighting the importance and challenge of improving educational provision in this context. Duflo et al. (2012) study a randomized experiment in which some schools were assigned to photographic monitoring and attendance-linked compensation. In treatment schools, a child photographed the teacher and other students at the beginning and end of each school day, and teachers' pay depended partly on their attendance. Their scheme reduced teacher absence by about 21 percentage points and improved test scores by about 0.17 standard deviations.

I reproduce the average treatment effect of 8.4 percentage points on recorded departure. In the extension, using 2,195 children in 111 schools, I find a statistically significant difference between the treatment effects for baseline writers and nonwriters. The treatment-control difference in recorded departure is 26.1 percentage points for students who took the written

baseline test and 5.0 points for those tested orally. The difference between these effects is 21.1 percentage points, with a school-clustered 95 percent confidence interval of 6.9 to 35.3 points. It remains statistically significant after Holm adjustment across the three departure outcomes ($p = .012$). My contribution is to extend the original analysis of average departure effects by looking at differences between these two baseline groups.

A growing literature examines the role that teacher monitoring and incentive schemes may play in improving educational outcomes. Kremer et al. (2005) document high teacher absence in government-run primary schools in India. A systematic review of impact evaluations by Guerrero et al. (2013) finds evidence that monitoring combined with incentives can increase teacher attendance, but weaker evidence of improvements in student learning. Cilliers et al. (2018) find that a program in Uganda providing financial incentives conditional on locally monitored attendance increased teacher attendance, whereas monitoring by itself was insufficient. Other papers focus on performance-based incentives for improving teacher effort or student learning without necessarily changing teacher attendance (Glewwe et al., 2010; Muralidharan & Sundararaman, 2011). These findings motivate looking beyond teacher attendance to examine children's subsequent participation.

Previous studies also suggest that effects of educational interventions on learning can be particularly large at the low end of the baseline distribution of test scores (e.g., Banerjee et al., 2007), and may depend on baseline knowledge and the organization of instruction (Duflo et al., 2011). Duflo et al. (2012) look at differences in effects on learning and school attendance according to whether children could take the baseline written test or had to be tested orally. They do not look at differences in leaving the center along this dimension. I use the same distinction as a rough measure of baseline preparation to examine whether the program's recorded-departure effect differs between these groups.

The original paper finds that treatment increased recorded departure from the centers, both overall and in cases where the recorded reason was entry to a government school. In my extension, the estimated difference between the treatment effects for writers and nonwriters is larger for residual departure than for the government-school marker, although the government-school estimate is imprecise. Residual departure includes children recorded as having left without a government-school marker, whether they had another recorded reason or no recorded reason. It does not tell us that they left education. Leaving a Seva Mandir center could reflect moving into government school or leaving for another reason. The data do not tell us what happened to every child.

These are exploratory results. The adjustment for multiple testing covers the three departure outcomes, but not the other extensions examined before choosing this one. The results do not tell us why the effects differ between writers and nonwriters. I offer some possible explanations, but there is little information in the data to tell them apart. There are also reasons to be careful about how departure is measured. The measure uses attendance at the latest up to five checks when the school was open. Treatment increased school opening, so it also changed the occasions on which children could be observed. Linden and Shastry (2012), in a different setting, also find that recorded attendance does not always match directly observed attendance. I therefore present these results with caution. They show a difference in recorded departure, but do not establish whether children left education or moved to another school.

This paper is organized as follows. Section 2 presents the replication results, Section 3 presents my extension, and Section 4 concludes.

2. Replication

Duflo et al. (2012) study whether camera monitoring and attendance-linked pay reduce teacher absenteeism and improve learning. They find about a 21-percentage-point reduction in teacher absence and a 0.17-standard-deviation gain in test scores. They run a field experiment with educational centers run by Seva Mandir in rural Udaipur, Rajasthan (Duflo et al., 2012). In these centers, which I will refer to as schools following the practice in Duflo et al. (2012), children learn basic Hindi and other things to prepare them for enrolling in government schools. Schools typically have a single teacher, so are closed when the teacher is not present. Duflo et al. (2012) randomize schools to a package of camera monitoring and attendance-linked pay; the treatment began in September 2003. I focus on results through February 2006, which was used to produce the two tables that I replicate; however, as of October 2009 the program was still active.

Duflo et al. (2012) randomize 120 schools, seven of which were excluded from the analysis following randomization but before the intervention was announced to schools, leaving 57 treatment and 56 control schools. In treatment schools, teachers and children were photographed at the beginning and end of each school day; researchers measured attendance through surprise visits in both treatment and control schools. Treatment schools' teachers were compensated based in part on attendance as indicated in the photos; control schools' teachers continued to receive a fixed wage.

Using data shared by Duflo et al. (2019) in a replication package (E112523V1), which consists of Stata files, I replicate the pooled all-teacher result in the paper’s original Table 2,

Panel A, and all three departure rows in its Table 10, and recreate these results and an extension (described below) in Python. Software and reproduction details appear in Appendix D. After completing this replication, I add the extension described below.

In the original Table 2, Panel A, the sample consists of 3,071 random visits to 113 schools. Duflo et al. (2012) estimate a regression of whether the school was open on a constant and treatment status and report standard errors clustered by school. To construct the data, I exclude a baseline period before treatment was announced, continue to exclude the same schools as Duflo et al. (2012), and categorize the data collected in late August as belonging to the intervention period of September 2003–February 2006.

I match the pooled result in the original Table 2, Panel A, of Duflo et al. (2012). In my reconstruction, schools were open in 78.7% of visits in treatment schools and 58.0% of visits in control schools. The estimated treatment effect, with SEs clustered at the school level in parentheses, is 0.207 (0.030). Rounding to match precision in the published table, I match these published values. Expressed in percentage points, the 95% confidence interval is [14.9, 26.6]. Table 1 reports both the attendance and departure replications.

In Table 10 of Duflo et al. (2012), the sample consists of 2,197 children in 111 schools. I match the results in Table 10, which includes regressions of departure indicators on a constant and treatment status and reports standard errors clustered by school. I use the algorithm in Duflo et al. (2012): for a given child, first select the latest up to five times they were checked when the school was open, from last to first. Next, recode whether or not they attended in those selected instances, and sum up these recodes. If the result is 0, then this child is classified as having departed under this rule. A child for which this is not true did not necessarily continue; this method of aggregation does not tell us that. The dependent variables are (1) recorded departure, (2) a government-school marker among recorded departures, and (3) residual departure, equal to recorded departure minus the marker.

I recreate this data on departure from the two raw data files. I merge these roster files on to school random checks using school id and visit date, and on to child enrollment and baseline writing records using school id × child id. To determine departure, I follow the procedure above. Some attendance codes lack documented meanings, and the reason fields differ in type. I follow the attendance recodes in Duflo et al. (2012) as well as their procedure for appending reason variables. In many instances the reason is blank; a blank does not itself classify a child as departed or supply a government-school marker. The marker equals one for a child classified as departed with reason “2” in at least one selected record. I divide departure up into an indicator for the recorded government-school marker and an indicator for residual

departure; these two outcomes sum to departure in my calculations. I chose the names of these categories carefully; the first is only a marker for whether the selected records contain a government-school reason marker, and not having this marker does not necessarily indicate leaving education. Appendix B describes this data more fully and discusses some inconsistencies between the numeric version of reasons children leave and a text version of these reasons.

At the reported level of precision, I match the coefficients and SEs in the original Table 10. Assignment to the bundled program causes a statistically significant 8.4-percentage-point increase in recorded departure (SE 4.1 points, p = .044), and a 10.1-percentage-point increase in the government-school marker (SE 3.5 points).

I next replicate the measure used in the original article. I find a −1.8-percentage-point change in residual departure (SE 3.1 points, p = .564). This calculation matches the child-level departure indicators provided in the replication package.

Table 1

Replication of Original Table 2, Panel A, and Table 10

Outcome	Original table	Published effect (SE)	Replicated effect (SE)	N
School open	2 Panel A	0.21 (0.03)	0.207 (0.030)	3,071
Recorded departure	10	0.08 (0.04)	0.084 (0.041)	2,197
Government-school marker	10	0.10 (0.03)	0.101 (0.035)	2,197
Residual departure	10	−0.02 (0.03)	−0.018 (0.031)	2,197

Note. Effects and standard errors are in probability units; 0.01 equals one percentage point. The attendance row is the pooled all-teacher result in original Table 2, Panel A (3,071 visits, 113 schools). The departure rows reproduce all three original Table 10 rows (2,197 children, 111 schools). Recorded departure means zero recoded attendance over the latest up to five open-school checks. The government-school marker requires recorded departure and reason “2” in a selected record; residual departure is the remainder. Standard errors are clustered by school. No significance stars are used. Source: Duflo et al. (2012) and the reconstructed Version 1 data.

3. Extension

The estimates below use the main sample described in Duflo et al. (2012). To facilitate matching to the study replication files, I drop two children with multiple baseline observations whose records disagree about the child's test status, resulting in 2,195 children from 111 schools. The sample includes 1,136 children from 57 treated schools and 1,059 children from 54 control schools. Without adding data or measures or imposing passing score cutoffs on test scores, there are 1,798 "nonwriters" and 397 "writers."

A motivation for studying departures is that the Seva Mandir-operated centers can be stepping-stones towards enrollment into government schools by teaching children foundational skills. One reason treatment may have had heterogeneous effects on departures may be that it caused children who were better prepared to enter government schools (with written baseline-test status used as a coarse preparation proxy) to be more likely to leave the centers. Of course, any heterogeneity in departures could be due to a variety of other reasons, including other differences between writers and nonwriters or differential impacts of monitoring on measurement of writer versus nonwriter attendance. Duflo et al. (2012) use this same distinction to analyze heterogeneous impacts of treatment on learning outcomes and attendance. This exercise examines whether the treatment effect on recorded departure differs between writers and nonwriters, with or without a government-school marker. Thus, these results should be viewed as exploratory, and do not estimate a treatment effect of ability or of writing.

I estimate

$$Y_{is} = \alpha + \tau\ T_s + \beta\ W_{is} + \delta(T_s \times W_{is}) + \varepsilon_{is}, \qquad (1)$$

where Y is the binary outcome being analyzed, T denotes treatment status of the child's school, and W denotes whether the child took the written test in the baseline ("writer") as opposed to oral test ("nonwriter"). I estimate a saturated version of this model, so the coefficients can be interpreted as the departure rate among nonwriters in control schools (α), nonwriters in treatment schools (α + τ), writers in control schools (α + β), and writers in treatment schools (α + β + τ + δ). Thus τ is the treatment effect for nonwriters, τ + δ is the treatment effect for writers, and δ is the difference in these two treatment effects. This specification puts equal weight on each child. School-clustered standard errors (with 110

degrees of freedom for critical t values) for the estimated treatment effect among writers account for covariance between τ and δ.

I estimate the model for three different outcomes in Y. Departed is the departure measure used by Duflo et al. (2012). Government School is an indicator equal to one if the child is classified as departed and has a government-school reason marker in at least one selected record. Residual Departure is Departed minus Government School. I conduct wild cluster bootstrap hypothesis tests that impose the null under test, use 9,999 school-level Rademacher wild-bootstrap-t draws (using a random number seed of 20260922), and account for the finite number of clusters following Cameron et al. (2008). The null sets $\delta = 0$ while leaving the treatment and writing main effects unrestricted. Table 3 reports p-values for δ, the writer-minus-nonwriter difference in treatment effects. I adjust the conventional school-clustered p-values across the three outcomes using Holm (1979), and report wild-bootstrap p-values separately. These results are exploratory as this correction does not account for the search discussed in Appendix C that considered a number of potential candidate exercises before turning to this analysis of heterogeneity.

As above, I view these results as exploratory and the characterization of baseline writers as better prepared as heuristic, not an empirical statement that these children were better students on average. Residual departure is an operational residual category, not a measure of educational dropout.

Table 2 reports results. Column 1 replicates the original Table 10 average recorded-departure coefficient. Columns 2–3 report group treatment effects from the interaction model. I leave out the two children with conflicting baseline writing records from the 2,197-child replication sample, leaving 2,195 for the interaction analysis. I report the estimates together with their sample sizes and school-clustered uncertainty. They use slightly different samples and answer different questions: the first column reports the treatment-control difference in departure, on average, while columns 2 and 3 report treatment-control differences by baseline written-test status. Figure 1 graphs the results of columns 2–3 and their direct difference.

Results indicate a 26.1 percentage point treatment-control difference in departure for baseline writers (“Writer” column) and a 5.0 percentage point difference for baseline nonwriters (“Nonwriter” column). These numbers should not be interpreted as before-and-after changes among the same children: they are treatment-control comparisons among different children. Nor can we interpret baseline nonwriters as unaffected by the program, as their confidence interval is wide enough to include small negative or greater than ten point effects. The interaction term shows that baseline writers had a 21.1 (95 percent CI: [6.9, 35.3])

percentage-point larger effect on departure from educational centers than did baseline nonwriters. The p-value on this interaction term is .004 using clustered standard errors and .007 using the wild-cluster bootstrap suggested in Cameron et al. (2008). With three departure outcomes tested, a Holm-adjusted p-value is .012.

Table 2

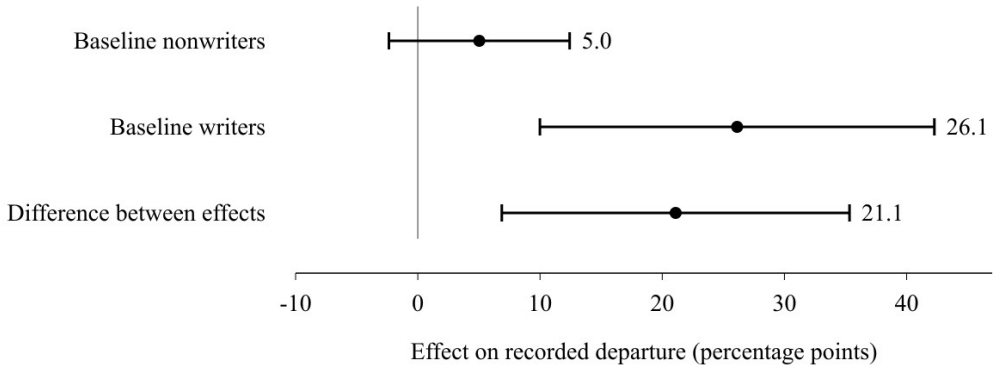
Extension of Original Table 10 by Baseline Writing Status

Recorded departure	**Average (1)**	**Nonwriter (2)**	**Writer (3)**
Treatment effect	8.4	5.0	26.1
Clustered SE	4.1	3.7	8.1
95% confidence interval	[0.2, 16.5]	[−2.4, 12.4]	[10.0, 42.3]
Children	2,197	1,798	397

Note. Effects, standard errors, and intervals are in percentage points. Column 1 repeats the aggregate 2,197-child replication in 111 schools. Columns 2–3 are τ and $\tau + \delta$ from equation (1), fitted to 2,195 children in 111 schools after excluding two conflicting baseline-writing records. Writer/nonwriter describes the child's baseline written-versus-oral test protocol. Covariance estimates are clustered by school; intervals use 110 degrees of freedom. The writer SE accounts for covariance between τ and δ. No significance stars are used.

Figure 1

Recorded Departure Effects and Their Difference by Baseline Writing Status



Note. Points show τ for baseline nonwriters, τ + δ for baseline writers, and δ for their direct difference from equation (1). Bars show 95% confidence intervals based on school-clustered SEs, with 110 degrees of freedom. All estimates use 2,195 children in 111 schools. Recorded departure follows the original last-up-to-five-open-check rule; it is not verified educational dropout. Units are percentage points. No significance stars are used.

The total-departure interaction does not establish why or where these children went when they left: was departure because they found a place in a government school or for some other reason? I next estimate the same model, with the interaction between school treatment and child baseline writing status on more refined outcomes in Table 3. The interaction with baseline writing status has a larger point estimate on Residual departure than on Government school, though I hesitate to make strong statements about these children's destinations based on one being classified as residual.

Results do not contradict those reported in the original study's Table 7, which found more days of instruction because schools were open more often. Panel C limits its analysis to children who stayed at the center and reports that presence was 14 percentage points higher for children who could only take the oral pretest and 10 points higher for those who could take the written test, a difference that is not significant. Nor do these results necessarily indicate that the children who benefited the most in learning (the original Table 9 reports larger estimated learning gains among children who took the written pretest) are the same children who transferred out, nor do they track what happened to individual children on all three margins (presence, learning, destination). My results in Table 2 extend the original study's Table 10 by splitting its reported dependent variable Leaving the center up by the groups in baseline tests that the original authors used in Tables 7 and 9.

Table 3 shows interaction results for total recorded departure, the government-school marker, and residual departure, under the definitions of departure from baseline school given above. I report all three to be clear about the limitation. Under these definitions, the government-marker and residual interaction point estimates sum to the total-departure interaction. The government school estimate does not provide evidence that the treatment effect on the recorded government-school marker was larger for writers than nonwriters, but the residual departure interaction estimate is precise, both without (p = .007) and with adjustments for multiple hypothesis testing. Departure without a government-school flag includes all other recorded departures, including those with missing destination reasons. The interpretation in the original paper of an increase in the move to government schools overall motivated this exercise, but the imprecise marker interaction and missing destination records do not establish whether transfers explain the difference between the groups.

Table 3

Extension of Original Table 10 Across Recorded Departure Outcomes

Outcome	δ	95% confidence interval	Clustered p	Holm p	Wild bootstrap p
Recorded departure	21.1	[6.9, 35.3]	.004	.012	.007
Government-school marker	4.8	[−10.5, 20.2]	.534	.534	—
Residual departure	16.3	[4.6, 27.9]	.007	.014	.010

Note. Table reports interaction coefficient δ from equation (1). Estimates and intervals are in percentage points. The sample is 2,195 children in 111 schools. Recorded departure follows the original latest-up-to-five-open-check rule. The government-school marker requires recorded departure and reason “2” in a selected record; residual departure is recorded departure minus this marker. Under the original coding, the government-marker and residual interaction point estimates sum to the total-departure interaction; their SEs, intervals, and p-values do not add. Conventional p-values and intervals use school-clustered SEs and 110 degrees of freedom. Holm adjusts the three conventional interaction p-values. Wild-bootstrap p-values use 9,999 null-imposed school-level Rademacher wild-bootstrap-t draws, seed 20260922, testing δ = 0 with unrestricted main effects. The dash means no government-marker bootstrap p-value was estimated/reported. No significance stars are used.

Results for departure are limited in important ways that should be kept in mind when considering their meaning. First, as discussed above, some children have fewer than five checks in which the baseline school is open, and some departure classifications involve undocumented attendance codes. Among the main sample used above, there are 145 with insufficient observations and 113 whose departure classifications involve undocumented attendance codes; 28 are in both groups, so that in total there are 230 with at least one of the problems. For consistency with the original paper I leave the observations coded as they were, but distinguish these from random sampling issues. Results controlling for existing school-block/geographic indicators and limiting the sample to end in December 2005 (to check dependence on the final January–February 2006 records, whose appended reason values become blank under the distributed Stata convention) are in Appendix A. Additional results excluding the 113 children with ambiguous attendance-based departure classifications and bounding the interaction effect under assumptions about departure among children with problems are also presented. Departure interaction estimates are positive in all cases, although exclusion changes the sample and the algebraic ranges hold the other observations fixed and are not confidence intervals. None of these solutions deal with the important limitations noted next.

Second and most important, because treatment increased the opening of schools, it affected measurement of whether children attended school. Departure is measured based on the last checks at which the school is open, and thus the last time at which it is possible the child would appear in the data. On average, schools were open for 15.4 (20.4) of the checks for baseline nonwriter children in control (treatment), and 15.2 (21.9) of checks for baseline writer children. Thus the last five times in which a school is open could occur in very different time periods, depending on whether the school is in treatment or control and the child's baseline writing status. Randomization still allows comparison of groups, but makes this far from an obvious interpretation as a treatment-independent measure of permanent educational exit. A child absent at all selected checks could have left education, moved, or just happened to not go on those days. A child who was present some days may or may not have been in school the entire time. It is unclear how much of the interaction effect above reflects changes in educational participation and how much the mechanics of measuring educational participation. Finally, reason fields are blank in 7,112 of 10,707 selected open-school check records (66.4%), which limits interpretation of the departure measure. In short, using the rules of departure given originally, the table provides evidence that treatment caused recorded departure from the baseline school to increase more for baseline writers than baseline nonwriters. More general implications for education are less clear.

4. Discussion and Conclusion

I show two complementary findings. First, I replicate the original study's effects on school opening and recorded departure. Second, the extension indicates that the effect on recorded departure was larger among children who took the written baseline test than among those who took the oral test. The treatment-control difference was 26.1 percentage points for baseline writers and 5.0 points for nonwriters. The difference between these effects was 21.1 percentage points, with a 95 percent confidence interval of 6.9 to 35.3 points. It remained statistically significant after Holm adjustment across the three departure outcomes.

The separate departure outcomes are an important qualification. The larger point estimate is for residual departure, while the government-school estimate is imprecise. This does not show that the intervention increased educational dropout among writers. Residual departure means that a child was recorded as having left the center without a government-school marker. It does not establish whether the child left education or moved to another school. Statistical significance also does not establish the process behind the difference between the two groups.

There are also limitations in the departure measure. It uses attendance at the latest up to five checks when the original school was open. Treatment increased school opening, so it also changed the opportunities and timing for observing children. Some children have fewer than five eligible checks, and some departure classifications depend on attendance codes whose meanings are not documented. My worry is not that randomization failed, but that recorded departure does not tell us whether a child permanently left education.

These results are exploratory. The Holm adjustment covers the three departure outcomes, but does not account for the other extensions considered before choosing this one. What the extension adds to the original paper is evidence that the effect on recorded departure differed depending on baseline writing status. The analysis does not establish why the effects differed or whether children's subsequent educational participation improved or worsened.

5. AI Use Statement

During May–October 2026, I used OpenAI Codex (GPT-5.6 Sol and Luna, GPT-6 Astra and Luna, and GPT-6.1 Sol) and Claude Code (Claude Opus 4.8, Opus 5, Fable 5, Fable 5.1, and Opus 5.5) to find information, help with coding and data analysis, and provide examples of structure and wording to guide my own writing. I used the tools for copyediting and formatting, and to help prepare the tables, figures, and appendices and check their content, relevance, and presentation against the seminar requirements. Records from multiple conversations with these tools have been preserved to document this use.

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Appendix A Fixed sensitivity analyses

Table A1

Fixed sensitivity analyses of the departure interaction

Specification	Interaction pp	95% CI pp	p	Children
Main model	21.1	[6.9, 35.3]	.004	2195
Geographic block controls	22.1	[8.2, 36.0]	.002	2195
December 2005 endpoint	19.7	[4.8, 34.6]	.010	2195
Exclude ambiguous codes	26.6	[12.0, 41.2]	< .001	2082

Note. Every model has 111 school clusters and reports the writer minus nonwriter contrast. Standard errors are clustered by school; 95 percent intervals and p-values use 110 degrees of freedom. The block model uses the existing geographic school block field, not established randomization strata. The calendar check truncates source dates but preserves the original observation rule. The exclusion changes the analyzed population. Sources are the fixed scope and sentinel sensitivity CSVs. No stars are used.

Table A1

retains the main interaction beside three fixed checks. Geographic controls use the existing school block field on the same 2,195-child sample. They are geographic indicators; the analysis does not establish that they are randomization strata. The coefficient is 22.1 percentage points, with interval 8.2 to 36.0 points. The calendar check applies the same last-open-check rule using dates through December 31, 2005. Its coefficient is 19.7 points, with interval 4.8 to 34.6 points. Neither check supplies a treatment-independent participation measure.

The attendance-code sensitivity removes all 113 ambiguous departure classifications, leaving 2,082 children in 111 schools. Its interaction is 26.6 points, with interval 12.0 to 41.2 points and clustered $p < .001$. The wild-bootstrap p-value is .001. The excluded cases differ by assignment and baseline status, so this estimate is a selection sensitivity. Its larger magnitude is not a reason to replace the primary estimate.

Table A2

Algebraic bounds for documented outcome ambiguity

Outcomes treated as unknown	Flagged children	Interaction range pp
Undocumented attendance codes	113	[16.3, 34.1]
Fewer than five open checks	145	[11.9, 36.4]
Either concern with overlap removed	230	[8.8, 45.6]

Note. Bounds assign any binary outcome to flagged children while retaining all other outcomes and cell denominators in the 2,195-child, 111-school sample. There are 28 overlapping flags. Ranges are writer minus nonwriter treatment-effect differences in percentage points. They are algebraic ranges with no standard errors or p-values, not confidence intervals. Sources are transition_incomplete_last5_bounds.csv and measurement/sentinel_interaction_bounds.json.

Table A2

treats each flagged binary outcome as unknown and assigns all permissible zeros and ones within each cell. It reports the smallest and largest resulting writer-minus-nonwriter contrast. The attendance-code range is 16.3 to 34.1 points. The incomplete-history range is 11.9 to 36.4 points. Their union gives the broader range of 8.8 to 45.6 points. These are algebraic identification ranges conditional on the retained sample and all other recorded outcomes. They have no standard errors or p-values and are not confidence intervals. They do not bound every possible error in the roster or the counterfactual observation process.

Appendix B Measurement and reconstruction details

The raw roster sources differ in the type of the departure-reason field. The main source stores it as text, while the appended source stores numbers. The distributed Stata append leaves the numeric source's reason values blank in the text column. The replication preserves that convention to match the original output. Mapping the raw numeric government-school code to its corresponding text marker changes 13 children's recorded destination classification in the 2,197-child replication cohort. It changes no total-departure indicators. This is a count comparison; the government-marker and residual interactions have not been re-estimated under that harmonized mapping. This is why total departure is the primary outcome and destinations remain secondary.

Across the selected rows in the consensus sample, 66.4 percent of reason fields are blank. Blanks constitute 69.9 percent in control and 63.3 percent in treatment. A blank is kept blank, and no unobserved destination is imputed. The residual measure is defined by absence of the recorded government-school marker within operational departure. It is not a direct observation of withdrawal from all education.

The original attendance recode maps several negative values and code 22 to zero. The package does not provide a codebook that establishes the meaning of each of those values. The sentinel audit flags operational departures with at least one such selected value. It identifies 28 control nonwriters, 11 control writers, 71 treatment nonwriters, and 3 treatment writers. These sum to 113. Of these children, 28 also have short observation histories. The remaining quality categories are 85 code-only cases, 117 history-only cases, and 1,965 children with neither documented flag.

The source sample remains fixed apart from the two conflicting baseline-writing records and the explicitly labeled exclusion check. No raw attendance, transfer-reason, or test-score observations are added. Calendar sensitivity and coding sensitivity are reported separately because they alter different parts of the measurement process. Neither is used to redefine the main result after seeing its significance.

Appendix C Prior candidate directions and selection

The original teacher-score attendance catch-up story is already explicit in the paper and is not treated as a new contribution. The candidate learning-by-teacher-score extension is more distinct, but its observed-test total interaction is 0.132 standard deviations per one-standard-deviation increase in baseline teacher score, with interval -0.009 to 0.273 and p = .067 in 1,625 source rows across 101 schools. Geographic controls reduce this interaction to 0.087, with interval -0.042 to 0.217. Those source rows represent 1,610 unique school-child keys. The result does not establish a difference at the conventional two-sided 5 percent level.

The fixed learning-domain family gives a nominal mathematics interaction p-value of .037, but Holm adjustment across mathematics, language, and total yields .110 for mathematics. The direct mathematics-minus-language interaction has p = .054, and the joint domain test has p = .090. These results do not justify selecting mathematics alone as an established new effect.

The learning-distribution route uses 1,737 observed source rows, 1,720 unique school-child keys, and 107 schools. Its fixed lower-quartile contrast is 0.081 standard deviations, with a school-bootstrap percentile interval from -0.081 to 0.219 and approximate two-sided p = .310. The median, upper-quartile, and interquartile-range contrasts also have intervals containing zero. These analyses are retained as legitimate but inconclusive alternatives. Their existence is part of the departure paper's exploratory search history.

Appendix D Reproducibility record

The source is Version 1 of replication package E112523V1 (Duflo et al., 2019). The original benchmark programs are Table1_Fig1_update.do, Table-2.do, and Table7_Table10.do in data/Incentives_FINAL/Programs. I reconstruct the selected results in Python and compare them with the published tables and distributed outputs. The supplied Stata programs are benchmarks; I did not execute them in Stata.

The raw departure reconstruction is in scripts/audit_transition_extension.py. The interaction model, clustered intervals, geographic controls, calendar endpoint check, and wild bootstrap are in scripts/audit_transition_scope.py. scripts/audit_sentinel_measurement.py generates the attendance-code flags and algebraic bounds; scripts/audit_transition_sentinels.py generates the exclusion sensitivity. The main bootstrap uses 9,999 school-level Rademacher draws and seed 20260922.

The numerical verification on October 15, 2026 used Python 3.13.4, NumPy 1.26.4, pandas 2.3.0, SciPy 1.15.3, and Matplotlib 3.10.3. It recalculated the attendance coefficient and clustered covariance from the original records, reconstructed Table 10 from raw data, checked the consensus-writing sample and interaction, and verified the saved 9,999-draw bootstrap p-values. All 12 checks passed. This verification does not claim a rerun of every original table, the structural model, or every alternative learning analysis.

The accompanying replication package includes the source data and benchmark programs, reconstruction scripts, saved statistical outputs, an exhibit generator, a README with reproduction commands, and a file-hash manifest. The main table values and coefficient plot are generated from the reconstructed results. Appendix C records the earlier candidate analyses; their stored outputs and scripts are included separately from the main departure analysis.

The paper and replication materials are available in the public GitHub repository.